## Research activity

**Checking components — 40 min.**

A component is chosen at random from a production coming from two workshops. Denote by $A$ the event "the component comes from the first workshop", $B = \overline{A}$ and $D$ the event "the component is defective". We are given $P(A) = 0.6$, $P_A(D) = 0.02$ and $P_B(D) = 0.05$. Approximate results will be rounded to $10^{-4}$.

1. Represent the situation by a weighted tree, then calculate $P(D)$.
2. The component chosen is defective. Calculate the probability that it comes from the first workshop.
3. We now choose twenty components independently, each following the previous model. Denote by $X$ the number of defective components among the twenty. Justify the distribution followed by $X$ and state its parameters.
4. Calculate $P(X = 0)$, $P(X \geq 2)$ and $E(X)$.
5. If the twenty components were taken from a single batch, which assumption of the model would need to be re-examined? Explain why.